Framework Selection Documentation

The sentiment analysis and review summarization system consists of two Transformer based models that are implemented by using the Hugging Face Transformers library. For sentiment classification, the project will use a pretrained BERT sequence classification model, while BART is used to generate summaries of customer reviews. Using a pretrained transformer model allows the project to leverage transfer learning, which reduces computational cost while maintaining strong performance based on specific tasks.

The sentiment classification component is implemented using the AutoTokenizer and AutoModelForSequenceClassification. The tokenizer converts raw customer reviews into token IDs and attention masks that are able to be processed by BERT. A three-class layer is added to the pretrained transformer, which produces probability scores for the three sentiment categories: negative, neutral, and positive. The model is fine-tuned using the Hugging Face Trainer API which manages the batching, optimization, validation, checkpointing, and metric computation throughout training.

The preprocessing pipeline begins by loading the Amazon Electronics review dataset and product metadata separately before merging them into the parent_asin product identifier. Reviews with missing text are removed, and excessive whitespace is cleaned from both titles and review bodies. Review titles and review text are then combined into a single input_text field that becomes the input to BERT. Then the customer star ratings are converted into sentiment categories, where ratings of one to two are labeled as negative, three as neutral, and four to five as positive These are then mapped to integer labels with negative being zero, neutral being 1, and positive being 2. In order to ensure balanced model training, this dataset is then divided into 70% training, 15%

validation, and 15% testing subsets using stratified sampling. This preserves the proportion of each sentiment across the datasets, preventing class imbalance from affecting model performance.

The BERT model is fine-tuned on the training dataset which validation performance is monitored after each training epoch. Dynamic padding is applied during training through DataCollatorWithPadding, reducing unnecessary computation by padding each batch only to the longest sequence within that batch rather than padding every review to a fixed and maximum length. Hyperparameters include the learning rate, batch size, number of epochs, warmup steps, and weight decay and are managed through a YAML configuration file, making the training pipeline easy to reproduce and modify. After training, the saved BERT model is evaluated on the unseen test dataset. Performance is measured using overall accuracy, Macro F1-score, Weighted F1-score, per-class F1-scores, a confusion matrix, and a classification report containing precision, recall, and F1-score for each sentiment category.

The review summarization component uses the pretrained facebook/bart-large-cnn model to generate abstractive summaries of Amazon product reviews. Reviews are grouped by product, and two summarization strategies are evaluated: a combined strategy, where positive, neutral, and negative reviews are summarized together, and a sentiment separated strategy, where reviews belonging to each sentiment category are summarized independently. Before summarization reviews are ranked according to their helpful votes so that the most informative customer feedback is prioritized as model input. Generated summaries are evaluated using both automatic and manual evaluation metrics. Automatic evaluation includes ROUGE-1, ROUGE-2, and ROUGE-L source coverage, lexical coverage, novelty ration, compression ratio, repetition ratio, and sentiment alignment using VADER sentiment analysis.